

LAVANYA SIMHAM

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SUMMARY

Full Stack Java Developer with **4+ years** of experience in **telecom** and **enterprise** environments, **building scalable** and **high-volume transactional systems** on AWS. **Experienced in developing backend services** using Java (8/11), Spring Boot, REST APIs, and Spring Security (OAuth2, JWT), along with **building responsive front-end features** using React.js and Angular that **integrate seamlessly with backend services**. **Skilled in working** with SQL and NoSQL databases, Apache Kafka, and Netflix Eureka, and **delivering reliable solutions** through Docker and Kubernetes based CI/CD pipelines on AWS (EC2, EKS) within **Agile (Scrum) SDLC** teams. Proven ability to collaborate with **cross-functional teams**, troubleshoot complex issues, maintain **code quality standards**, and deliver reliable, high-performance applications. Committed to **continuous learning** and **building innovative solutions** that enhance user experience and business outcomes.

PROFESSIONAL EXPERIENCE

Full Stack Java Developer

India

Tata Consultancy Services

May 2023 – Dec 2023

- Architected **Java 11, Spring Boot microservices** using **Spring Data JPA and Hibernate**, supporting **18+ order lifecycle workflows** and processing **120K+ daily telecom transactions** in production environments.
- Developed **responsive UI components** using **React.js**, integrating frontend workflows with **RESTful APIs**, eliminating manual order tracking for **6 fulfillment teams** and enabling **real-time operational visibility**.
- **Applied** backend security using **Spring Security, OAuth2, and JWT**, enforcing role-based access control across **14 internal and partner-facing APIs** supporting enterprise telecom integrations.
- Streamlined **event-driven architecture** with **Apache Kafka and Netflix Eureka**, enabling real-time status propagation across **9 downstream systems** with **sub-3-second message delivery latency**.
- Participated in peer code reviews, unit testing (JUnit, Mockito), integration validation, and Agile sprint ceremonies to ensure production-ready deliverables.
- Automated CI/CD pipelines using Jenkins, Docker, and Kubernetes on **AWS EC2/EKS**, managing **15+ cloud deployments** with **S3-backed artifact storage** and reducing release execution time to under **25 minutes**.

Full Stack Java Developer

India

Brillio

May 2022 – April 2023

- Designed **Spring Boot microservices, Java 8, JDBC, and MySQL-backed services** for IP lifecycle management, automating allocation and reclamation across **60K+ network IP records** supporting **enterprise telecom operations**.
- Implemented **RESTful APIs, Spring MVC, and Swagger / OpenAPI** documentation standards, enabling **12+ external system integrations** and accelerating **client onboarding timelines**.
- Constructed **React-based UI components from Figma designs**, delivering real-time IP visibility dashboards used daily by **5 network operations teams** to monitor allocation status and operational health.
- Optimized **Spring Data JPA, Hibernate**, indexed queries, and stored procedures, reducing IP lookup latency to **2.1 seconds** across **high-volume allocation workflows**.
- Hardened application security using **Spring Security, JWT, and Veracode** scans, resolving **40+ security findings** prior to production deployment in **regulated environments**.
- Deployed Spring Boot services on **AWS** using **EC2 and RDS**, coordinating application configuration, secure connectivity, and **Lambda-triggered operational tasks** for REST-based integrations across **12+ enterprise systems**.
- Contributed to structured SDLC processes including architecture documentation, code reviews, automated testing, and continuous integration workflows.

Software Developer

India

Infosys

January 2021 – May 2022

- Built end-to-end backend features using **Java, Spring MVC, Hibernate, and RESTful APIs**, delivering **8 production modules** with clearly separated **service and data layers**.
- Designed relational database schemas using **Oracle SQL and MySQL**, defining tables and constraints, supporting **transactional workflows** across **4 finance and operations systems**.
- Delivered **user-facing operational workflows** enabling daily usage by **600+ internal and business users**, supporting finance and operations teams across **4 enterprise systems**.
- Managed source control using **Git and GitLab**, resolving merge conflicts and enforcing branch hygiene, coordinating code integration across **6-person Agile teams** during sprint and release cycles.
- Implemented unit and integration testing using **JUnit, Mockito, and Postman** to validate 90+ REST endpoints prior to release.
- Troubleshooted and resolved application defects across staging and production environments, improving system stability and user

satisfaction.

- Created technical design documents and API documentation to support QA validation and downstream system integrations.
- Collaborated in **Agile teams**, participating in peer design reviews, backlog refinement, and sprint delivery cycles.

Software Developer

India

Robosoft Technologies

November 2019 – December 2020

- Established backend service components supporting **7 internal enterprise applications**, implementing **Java**, **Spring MVC**, and **REST APIs** in alignment with predefined architectural guidelines.
- Maintained relational data access logic across **3 finance and operations platforms**, designing optimized queries and joins using **Oracle SQL** and **MySQL** for transactional workflows.
- Assembled form-driven operational workflows accessed by **1,200+ internal users**, supporting **high-frequency daily transaction processing** across multiple enterprise applications.
- Managed source code workflows across **6 developers**, coordinating branching and merges using **Git** and **GitLab** to support consistent Agile sprint delivery cycles.
- Executed unit and API testing validating **90+ REST endpoints**, applying **JUnit**, **Mockito**, and **Postman** to verify functional correctness before staging promotion.
- Collaborated with **QA**, **Product Owners**, and **DevOps** teams throughout SDLC execution, contributing code changes across **11 production releases** delivered on schedule.

TECHNICAL SKILLS

- **Programming Languages:** Java (8/11/17), Java/J2EE, Python, SQL, JavaScript (ES6+), TypeScript
- **Backend Technologies:** Spring, Spring Boot, Spring MVC, Spring Security (OAuth2, JWT), Spring Data JPA, Hibernate, RESTful APIs, JDBC, Servlets
- **Frontend Technologies:** React.js, Angular, HTML5, CSS3
- **Databases:** MySQL, PostgreSQL, Microsoft SQL Server, Oracle, MongoDB
- **Messaging & Event Streaming:** Apache Kafka
- **Cloud Platforms:** AWS (EC2, EKS, S3, RDS)
- **Containerization & Orchestration:** Docker, Kubernetes (AWS EKS)
- **CI/CD & DevOps:** Jenkins, GitHub Actions, Maven, Gradle, CI/CD Pipelines, Automated Deployments
- **Version Control:** Git, GitHub, GitLab
- **Testing & Quality:** JUnit, Mockito, Postman, SonarQube, JaCoCo, Jest, Cypress
- **API & Documentation:** Swagger / OpenAPI
- **Monitoring & Logging:** Splunk, CloudWatch
- **Development Tools:** IntelliJ IDEA, Eclipse, Spring Tool Suite (STS), Visual Studio Code
- **Operating Systems:** Linux, Unix
- **Methodologies:** Agile (Scrum), SDLC, SOLID principles, Test-Driven Development (TDD)
- **Architecture:** Monolithic, Microservices, MVC Architecture, Event-Driven Architecture.

PROJECTS

SmartEats – Cloud-Native Food Ordering Platform [Link](#)

Tech Stack : Java, Spring Boot, Angular, Eureka, Docker, Kubernetes (AWS EKS), AWS RDS (MySQL), MongoDB, Jenkins, ArgoCD

- Completed a production-style food ordering platform using **Java**, **Spring Boot microservices**, and **Angular**, enabling **end-to-end order placement, tracking, and history storage** across multiple service boundaries.
- Implemented service deployment on **AWS EKS** with containerized **Spring Boot** microservices, configuring namespaces, service discovery, and ingress routing to support scalable order processing across multiple environments.
- Operationalized cloud deployment using **Docker**, **Kubernetes** (AWS EKS), **Jenkins**, and **ArgoCD**, establishing **CI/CD workflows**, service discovery with Eureka, and scalable traffic routing via ALB Ingress.

EDUCATION

Missouri State University

MO, USA

Master of Science in Computer Science

January 2024 – December 2025

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** – Amazon Web Services